

Matrix Decompositions

Mathematical Models For Data Science

Pratyush Dayal

Indian Institute of Technology Gandhinagar

Why and what?

Analogy

- Factoring of numbers, such as the factoring of 21 into prime numbers $7 \cdot 3$.
- Also called **Matrix Factorization**.
- Different representation using factors of interpretable matrices. For example:
 - Cholesky decomposition: Square-root-like operation for symmetric, positive definite matrices
 - Singular value decomposition: Extends this factorization to non-square matrices
- Represent very large numerical data in convenient form for analysis.

Determinant and Trace

Characteristic Polynomial

For $\lambda \in \mathbb{R}$ and a square matrix $A \in \mathbb{R}^{n \times n}$,

$$p_A(\lambda) := \det(A - \lambda I)$$

is the characteristic polynomial of A , expressed as

$$p_A(\lambda) = c_0 + c_1\lambda + c_2\lambda^2 + \dots + c_{n-1}\lambda^{n-1} + (-1)^n\lambda^n$$

where $c_0, \dots, c_{n-1}$ are coefficients in $\mathbb{R}$. Specifically,

$$c_0 = \det(A)$$

$$c_{n-1} = (-1)^{n-1} \text{tr}(A)$$

The characteristic polynomial allows computation of eigenvalues and eigenvectors

- **Collinearity and Codirection:** Two vectors that point in the same direction are called codirected. Two vectors are collinear if they codirected point in the same or the opposite direction.
- Let $\mathbf{x}$ be an eigenvector of A associated with eigenvalue λ . Then for any $c \in \mathbb{R} \setminus \{0\}$, it holds that $c\mathbf{x}$ is also an eigenvector of A associated with eigenvalue λ . This can be shown as follows:

$$A(\mathbf{c}\mathbf{x}) = A(c\mathbf{x}) = cA\mathbf{x} = c\lambda\mathbf{x} = \lambda(c\mathbf{x}). \quad (4.4)$$

Thus, all vectors that are collinear to $\mathbf{x}$ (i.e., vectors of the form $c\mathbf{x}$, where $c \in \mathbb{R} \setminus \{0\}$) are also eigenvectors of A associated with eigenvalue λ .

